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COURSE BASED PROJECT

Smart Rover: A Multi-Function Robotic Car

(Arduino based Obstacle Avoiding, Bluetooth, terminal and Voice Control Robot Car)

SCHOOL OF ELECTRONICS ENGINEERING

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Abstract

The multi-function robotic car project presents a versatile Arduino-based vehicle capable of operating in four distinct modes: obstacle avoiding, Bluetooth control, terminal control, and voice control. By integrating sensors, wireless modules, and software logic, the car autonomously navigates obstacles, responds to commands from a smartphone, accepts text-based instructions via a serial terminal, and interprets voice commands. This project showcases the practical implementation of automation, wireless communication, and speech recognition, serving as an educational platform for embedded systems and robotics.

Introduction

Robotics has revolutionized automation and smart systems. Enhancing functionality with multiple control modes increases adaptability. This project introduces a robotic car built on Arduino technology, operating in four modes: obstacle avoiding, Bluetooth control, terminal control, and voice control. The system provides both autonomous and manual operation, making it suitable for varied environments and practical learning in electronics and programming.

An obstacle avoiding robot is a fully autonomous robot which can avoid any obstacles which it comes across, when it is moving forward. When the robot meets an obstacle while moving forward, it automatically stops moving forward and makes a step back. Then it slightly time and again goes forward, avoiding the obstacle.

It's also great for students to learn robotics and hands-on STEM skills. By assembling the hardware, connecting sensors, writing code, and fixing errors, we learned how real robot systems work. Obstacle-avoiding robots are an important part of modern robotics. They use sensors to detect barriers and move around them on their own. Adding Bluetooth and voice control makes the robot even more flexible. Projects like this help students understand how automation, electronics, and programming work together, and can be used for useful tasks in real life.

II. Literature Review

Classic obstacle-avoiding robots are limited by their reliance on pre-set logistics, typically using infrared or ultrasonic sensors for obstacle detection and avoidance. Recent studies highlight improvements with sensor fusion, machine learning, and advanced controllers, but many designs lack flexible, interactive interfaces for direct user input. Contemporary approaches focus on modularity and wireless technology—Bluetooth and voice recognition—improving human-robot interaction. Many robots have been made before that use sensors to avoid obstacles or can be controlled through Bluetooth or voice commands. Ultrasonic sensors are common for obstacle avoiding, while Bluetooth modules connect robots to

smartphones for easy control. Voice-controlled robots are now more popular with the use of voice recognition modules or apps. Combining these modes gives the robot more flexibility and real-world applications, as shown by various research papers and online tutorials.

III. Problem Statement

Traditional robots programmed solely for autonomous navigation cannot adapt to dynamic user requirements or changing environments. The lack of wireless and voice interfaces reduces their effectiveness in collaborative and interactive settings. Controlling and automating robot movements is often limited by just one method, making robots less useful for different needs and situations. There is a need for a single robot that can switch between autonomous obstacle avoiding and multiple manual controls like Bluetooth, terminal input, and voice commands—all in one system.

IV. Proposed Methodology

The robotic car is designed to operate in four modes—Obstacle Avoiding, Bluetooth Control, Terminal Control, and Voice Control—using an Arduino Uno as the central controller. Each mode is activated by a toggle switch or software command, enabling flexible operation. Our robot augments the existing design with Bluetooth and voice control modules. Through Bluetooth, users may remotely drive the robot with a smartphone or computer, enjoying wireless convenience. Voice control adds hands-free interactivity for commanding the robot using simple spoken instructions. Together, these features provide a comprehensive, adaptable, and engaging robotic car platform for practical and educational use.

The design enhances standard robotic cars as follows:

Hardware Implementation :

- **Microcontroller:** Arduino Uno manages inputs, outputs, and logic for all modes.
- **Motors & Driver:** Four DC motors for movement, controlled by an L298N motor driver module.
- **Sensors:**
 - **Ultrasonic Sensor (HC-SR04):** Measures distance to obstacles in front of the robot for autonomous navigation.
- **Communication Modules:**
 - **Bluetooth Module (HC-05/06):** Receives remote commands from a smartphone application.
 - **Voice Recognition Module OR phone app:** Interprets basic voice commands for movement.
- **User Interface:** Toggle switch or mobile app to select desired mode.
- **Power Supply:** Rechargeable battery for portability.

Software Development :

- **Programming:** Written in Arduino IDE using C/C++.
- **Mode Handling:** The main program loop checks the selected mode and executes its logic:

- **Obstacle Avoiding Mode:** Continuously reads the ultrasonic sensor. If an object is detected within a threshold distance, the robot turns or reverses to avoid.
- **Bluetooth Control Mode:** Parses incoming serial data from the Bluetooth module to control movement (forward, backward, left, right, stop).
- **Terminal Control Mode:** Accepts and executes movement commands sent via a serial interface from a PC or phone..

Algorithm :

1. On power-up, system initializes all components and connections.
2. The main loop prioritizes sensor-based obstacle avoidance. If an obstacle is detected within a threshold, the car stops, reverses, or turns.
3. The robot listens continuously for Bluetooth commands—like ‘F’, ‘B’, ‘L’, ‘R’—which override autonomous operation.
4. Voice commands are processed via a mobile app and then translated into Bluetooth signals, offering hands-free control.
5. Manual and autonomous modes can run concurrently, with user input given immediate precedence.

Testing: Modular testing and debugging ensure response consistency to obstacles and wireless commands. Optimizing sensor placement and code logic tunes real-world performance for speed, battery usage, and obstacle handling.

V. Results and Discussion

The complete robotic car system delivered strong results across all four modes:

- **Obstacle Avoiding Mode:**
The car successfully navigated both simple and complex environments, detecting and avoiding obstacles using the ultrasonic sensor. It was able to reroute itself accurately and prevent collisions in real time, demonstrating reliable autonomous operation.
- **Bluetooth Control Mode:**
Bluetooth mode allowed users to precisely control the car’s movements using a smartphone app within standard Bluetooth range. This was especially useful for indoor demonstrations and offered a flexible, interactive user experience.
- **Voice Control Mode:**
Voice commands enabled hands-free operation of the robot. The system responded well to clear and distinct commands, though recognition accuracy decreased in noisy environments. This mode demonstrated the value and challenges of speech interfaces in robotics.
- **Terminal Control Mode:**
Through a serial terminal (via USB or Bluetooth), users could send direct commands to the robot. This feature provided an additional layer of manual control and was useful for testing, debugging, or operating the robot in environments where app-based control was less practical.

VI. Conclusion

The project successfully demonstrates a cost-effective, open-source robotic car that unifies

autonomous navigation, Bluetooth wireless control, and hands-free voice commands on a single Arduino platform. Its success in both educational and prototype domains highlights the relevance of modular and extensible robotics in today's technological landscape, laying a foundation for further innovation.

This table gives information about what are limitations of this robot and how to overcome it.

Limitation in Existing System	How It Can Be Overcome
Only simple avoidance—may stop at obstacle rather than reroute	Use more advanced logic to turn/manoeuvre around obstacles
Bluetooth range is limited	Employ Wi-Fi/Bluetooth mesh or robust signal amplifiers
Basic voice commands, limited reliability in noisy environments	Integrate offline/onsite voice recognition or filters
No feedback on environment or robot status to user	Add sensors/LCD/voice output for live feedback
Relies only on single sensor position	Use sensor arrays for 360° obstacle detection
Limited battery power runs out quickly	Use power management and rechargeable batteries

VII. Future Scope

Possible future enhancements:

- Incorporate Wi-Fi or mesh networking for greater range and multi-user/model control
- Expand sensor arrays for 360° awareness and environmental mapping
- Implement offline or on-device voice recognition for reliability in remote/industrial settings
- Add visual (camera) modules and feedback (LCD displays, voice feedback)
- Integrate advanced algorithms for smart path planning and obstacle intelligence
- Develop rechargeable battery management with solar options or real-time power monitoring

VIII. References

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